

OBJECT ORIENTED PROGRAMING

OOP

PRACTICALS

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SECTION: B

ROLL NO: CT-25064

LAB NO: 05(ADDITIONAL TASK)

COURSE CODE: CT-260

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ADDITIONAL TASK:

✓ DIFFERENCE BETWEEN MULTILEVEL AND MULTIPLE INHERITANCE.

Feature	Multilevel Inheritance	Multiple Inheritance
Definition	▪ A class inherits from a class which itself inherits from another class	▪ A class inherits from more than one base class at the same time
Structure	▪ $A \rightarrow B \rightarrow C$ (chain form)	▪ $A + B \rightarrow C$ (one child, many parents)
Number of Parent Classes	▪ Only one direct parent at each level	▪ Two or more parent classes
Reusability Flow	▪ Step by step, level wise	▪ From multiple sources simultaneously
Complexity	▪ Simple and linear	▪ Slightly complex due to ambiguity problem



FUNCTIONS:

Multilevel Inheritance

Functions:

1. **Stepwise Feature Extension**
Allows functionality to be extended level by level (Base → Derived → Further Derived).
 2. **Code Reusability Across Levels**
A lower-level class can reuse properties of all upper-level classes.
 3. **Logical Hierarchy Representation**
Helps model real-world layered relationships like
`Person → Student → GraduateStudent.`
 4. **Progressive Specialization**
Each new class can add more specific behavior without modifying earlier classes.
 5. **Improves Maintainability**
Changes in base class automatically reflect in all derived levels.
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Multiple Inheritance

Functions:

1. **Combining Features of Multiple Classes**
A single class can use functionality from more than one base class.
2. **Code Reusability from Different Sources**
Avoids rewriting similar features in a new class.
3. **Supports Hybrid Behavior**
Useful when an object needs characteristics of different types
Example: `Teacher + Researcher → Professor.`
4. **Reduces Code Duplication**
Common functionality can be placed in separate base classes.
5. **Improves Design Flexibility**
Allows modular design where responsibilities are separated into different classes.



BASIC PROGRAMS:

MULTILEVEL INHERITANCE:

```
#include <iostream>
using namespace std;

// Base class
class Grandparent {
public:
    void familyName() {
        cout << "Family Name: Khan" << endl;
    }
};

// Derived from Grandparent
class Parent : public Grandparent {
public:
    void profession() {
        cout << "Profession: Doctor" << endl;
    }
};

// Derived from Parent
class Child : public Parent {
public:
    void school() {
        cout << "School: ABC School" << endl;
    }
};

int main() {
```

```
    Child obj;
    obj.familyName(); // From Grandparent
    obj.profession(); // From Parent
    obj.school();    // From Child
}
```

MULTIPLE INHERITANCE:

```
#include <iostream>
using namespace std;
```

```
// First base class
class Father {
public:
    void height() {
        cout << "Height: Tall" << endl;
    }
};
```

```
// Second base class
class Mother {
public:
    void eyes() {
        cout << "Eyes: Brown" << endl;
    }
};
```

```
// Derived from both Father and Mother
class Child : public Father, public Mother {
public:
    void hobby() {
        cout << "Hobby: Painting" << endl;
    }
};
```

```
int main() {  
    Child obj;  
    obj.height(); // From Father  
    obj.eyes();   // From Mother  
    obj.hobby();  // Own function  
}
```